***PRACTICAL – 2***

# *Implementation and Time Analysis of Linear and Binary Search Algorithm*

## *Objective:*

The objective of this practical is to implement ***Linear Search*** and ***Binary Search*** algorithms and compare their performance based on ***time complexity***, ***space complexity***, and ***execution time***. This helps in understanding the efficiency of both searching techniques and selecting the appropriate algorithm for different applications.

## *1.Linear Search*

### *Algorithm:*

START

INPUT A
INPUT X

FOR i ← 0 TO len(A) - 1
IF A[i] == X THEN

PRINT "Element found at index", i
EXIT
END IF
END FOR

PRINT "Element not found"

STOP

### *PythonCode:*

```
import time
# Linear Search Function
def linear_search(arr, n, key):
    for i in range(n):
        if arr[i] == key:
            return i
    return -1
# Take input from the user
n = int(input("Enter the number of elements: "))
arr = []
print("Enter the elements:")
for i in range(n):
    arr.append(int(input()))
key = int(input("Enter the element to search: "))
# Measure execution time
start_time = time.perf_counter()
result = linear_search(arr, n, key)
```

```
end_time = time.perf_counter()
# Display result
if result != -1:
    print("\nElement found at position:", result + 1)
else:
    print("\nElement not found")
# Display execution time
execution_time = end_time - start_time
print("Execution Time: {:.10f} seconds".format(execution_time))

 #Enter your EN.No
 print("EN.No:92460118683")
```

## *OUTPUT:*

```
PS C:\Users\GOUTHAM\OneDrive\Desktop\Daa practical 2> & C:\Users\GOUTHAM\AppData\Local\Python\pythoncore-3
.14-64\python.exe "c:/Users/GOUTHAM/OneDrive/Desktop/Daa practical 2/Linearsearch.py"
Enter the number of elements: 6
Enter 6 elements:
9
5
7
6
1
2
Enter the element to search: 7
Element found at position: 3

Time Complexity:
Best Case: O(1)
Average Case: O(n)
Worst Case: O(n)
Space Complexity: O(1)

Enrollment Number: 92460118683
PS C:\Users\GOUTHAM\OneDrive\Desktop\Daa practical 2>
```

***Time Complexity:***

| *Case* | *Time Complexity* | *Description* |
|---|---|---|
| *Best Case* | *O(1)* | *Element is at the first position.* |
| *Average Case* | *O(n)* | *Element is in the middle or random position.* |
| *Worst case* | *O(n)* | *Element is at the last position or absent.* |

***Space Complexity:***

| ***Space Complexity*** | ***Description*** |
|---|---|
| ***O(1)*** | ***Uses only a few variables; no extra memory is required.*** |

## *2.Binary Search*

***Algorithm:***

START

INPUT sorted list A

INPUT X

low ← 0

high ← len(A)-1

WHILE low <=high

mid ← (low+high)//2

IF A[mid]==XTHEN

PRINT"Elementfoundatindex",mid

EXIT

ELSEIFA[mid]<XTHEN

low←mid+1

ELSE

high←mid-1

***E***ND IF

ENDWHILE

PRINT "Element not found"

STOP

***Python Code***

```
import time
# Binary Search Function
def binary_search(arr, n, key):
    low = 0
    high = n - 1
    while low <= high:
```

```
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1
# Take input from the user
n = int(input("Enter the number of elements: "))
arr = []
print("Enter the elements:")
for i in range(n):
    arr.append(int(input()))
key = int(input("Enter the element to search: "))
# Sort the array
arr.sort()
# Measure execution time
start_time = time.perf_counter()
result = binary_search(arr, n, key)
end_time = time.perf_counter()
# Display sorted array
print("\nSorted Array:", arr)
# Display result
if result != -1:
    print("Element found at position:", result + 1)
else:
    print("Element not found")

# Display execution time
execution_time = end_time - start_time
print("Execution Time: {:.10f} seconds".format(execution_time))

#Enter your EN.No
print("EN.No:92460118683")
```

## *OUTPUT:*

```
PS C:\Users\GOUTHAM\OneDrive\Desktop\Daa practical 2> & C:\Users\GOUTHAM\AppData\Local\Python\pythoncore-3.14-64\python
.exe "c:/Users/GOUTHAM/OneDrive/Desktop/Daa practical 2/Binarysearch.py"
Enter the number of elements: 5
Enter 5 elements in sorted order:
7
8
5
4
1
Enter the element to search: 2
Element not found

Time Complexity:
Best Case: O(1)
Average Case: O(log n)
Worst Case: O(log n)
Space Complexity: O(1)

Enrollment Number: 92460118683
PS C:\Users\GOUTHAM\OneDrive\Desktop\Daa practical 2>
```

***Time Complexity:***

| *Case* | *Time Complexity* | *Description* |
|---|---|---|
| *Best Case* | *O(1)* | *Element is found at the middle.* |
| *Average Case* | *O(log n)* | *Search space is halved repeatedly.* |
| *Worst case* | *O(log n)* | *Maximum divisions before finding or rejecting the element.* |

## *Space Complexity:*

| *Space Complexity* | *Description* |
|---|---|
| *O(1)* | *Uses only a few variables (low, high, mid); no extra memory is required.* |